An AI-Powered Sound Design Assistant for Serum 2

Serum Synthesis Guide

Colin Busak

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SERUM SYNTHESIS GUIDE

AN AI-POWERED SOUND DESIGN ASSISTANT FOR SERUM 2

Overview

The Serum Synthesis Guide is an AI-powered assistant designed to help music producers learn and experiment with modular synthesis within Xfer's Serum 2 interface. This tool can generate fully functional synth patches within Serum 2 based on the user's input, guidance, and feedback. For every patch that it generates, it also provides the user with step-by-step instructions on how to recreate the patch in Serum 2, including all parameters, effects, and modulations.

With that said, the tool's primary function is to provide educational guidance as to how and why each synth patch works and sounds the way it does, as well as tips on how to incorporate your creative vision into your music production workflow.

An Educational and Ethical AI Tool

The Serum Synthesis Guide seeks to overcome two major accessibility barriers. Primarily, it is designed for music producers with an intermediate level of experience in modular synthesis, and music production in general, that are looking to expand their knowledge and expertise. It assumes that its user has a solid grasp on the fundamentals of these concepts and deliberately attempts to build on top of that foundation. Modular synthesis is a vastly complex and variable musical tool. For less experienced producers, it can be an overwhelming task to dive into its more intricate capabilities. The guide bridges this gap in an interactive, creative way that augments the user's music production workflow.

The second accessibility barrier the guide seeks to overcome is the ethical use of AI in a creative field. Generative AI has been and continues to be used unethically in the music industry to create entire songs, albums, artists, and bands. It often feeds off real artists and their work as the foundation to create soulless bodies of work that, surprisingly, have begun to rise through the industry and onto billboard charts. Aside from the lack of creativity and human touch, these AI-generated creations steal meaningful space and spotlight away from real musicians in an already heavily saturated industry. Additionally, studies have shown that generative AI utilizes more resources than any other form of AI, leaving behind a permanent, negative environmental footprint.

The Serum Synthesis Guide subverts this notion, functioning as a learning tool and creative partner, not a replacement for your artistic vision or workflow. It is dialogue-driven, making human input and feedback a crucial pillar of its design. Additionally, its sole reference for its outputs is the Serum 2 User Guide, meaning it won't pull from any content on the internet nor other established artists or their work.

Design and Goals

When I set out to create this prototype, I too was, and still am, in the process of learning modular synthesis through Serum 2. This idea actually stemmed from a previous project of mine, in which I created an AI chatbot using BoodleBox that would provide guidance and creative feedback for my music production workflow. The scope of this project was significantly broader. the chatbot operated around the user's initial input, which could be anything from a loop idea to an almost fully realized song. It would then attempt to build off this input, providing whatever musical or aesthetic guidance it could to help the user reach the end goal of their idea, not unlike the Serum Synthesis Guide. Figure 1 below shows this chatbot's interface and a sample prompt:

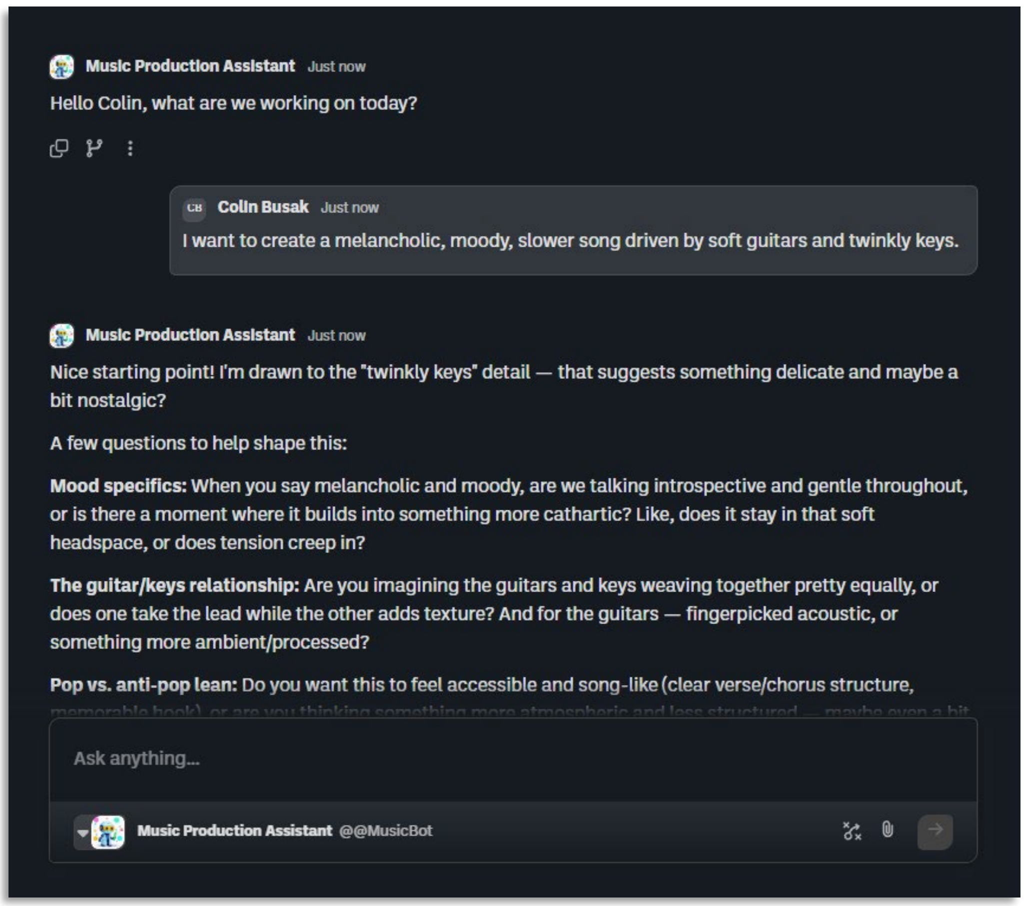


Figure 1: The Music Production Assistant's interface and sample prompt

My goal with that chatbot was to explore a more ethical approach to AI in a creative field, which became the working theme for the remainder of my projects in TCID 4060. After completing this project, I knew I wanted to dive deeper into this idea, while also narrowing the overall scope, purpose, and target user. Ultimately, this fostered the seeds of what would become the Serum Synthesis Guide: a user-driven AI tool and expert modular synthesis assistant.

The tool initially began as another chatbot. However, as the idea continued to develop and become more robust, I realized that the potential of this tool extended far beyond a simple chatbot interface. I also realized that making the tool accessible exclusively through a chatbot would likely turn many users in its target demographic, (those with justified hesitations regarding AI) away. To pivot from this design, I decided to transition the idea from a chatbot to a fully standalone HTML file. This file is accessible to anyone through any browser. The current prototype website is more limited in functionality compared to the chatbot but showcases the potential of this idea in a more rhetorically appropriate format.

I believe this design could be pushed even further and would function most optimally as a standalone VST (Virtual Studio Technology) plugin that could be used in any DAW (Digital Audio Workstation) software. However, given the time constraints and the limitations of my own technological knowledge, this was simply unfeasible for the purpose of this project and prototype.

Process of Making Prototype

Phase.7¿Initial.Configuration.and.Testing

As mentioned previously, I began the development process of this prototype in BoodleBox, a software that allows users to design AI-powered chatbots within a streamlined interface. I picked this platform primarily out of familiarity; I had used BoodleBox to create the Music Production Assistant and found it to be a useful tool for starting ideas. The chatbot builder also contains a "Vibe Bot Builder" assistant for developing chatbots. This assistant essentially functions as an entirely separate chatbot in which users can "vibe" code the instructions and general configuration for their desired chatbot using AI prompting.

Using the Vibe Bot Builder, I created the initial idea and configuration for what would become the Serum Synthesis Guide using the following prompt shown in Figure 2:

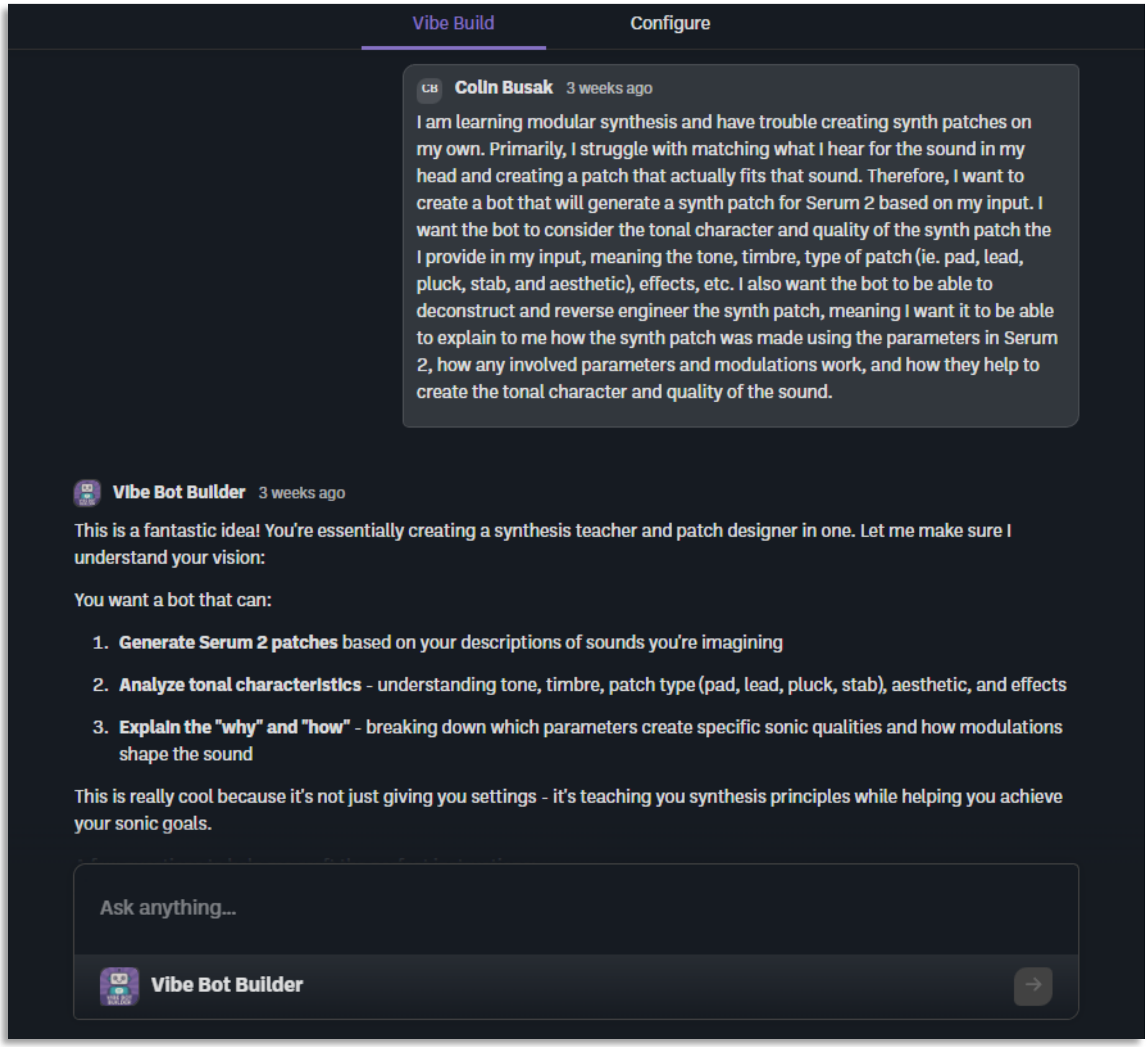


Figure 2: Initial prompt for Vibe Bot Builder and response

From here, I was able to provide more context as to how the chatbot should function, respond, feel from a user perspective, and assume from a knowledge base. I had already learned a lot from my previous chatbot project and knew what kinds of prompts I needed to provide the Vibe Bot Builder to accomplish my goal.

My prompts were both concise and explicit in what I wanted and like the bot I was designing, I also provided step-by step instructions for the Vibe Bot Builder:

- When explaining patches, I want it to explain synthesis fundamentals.
- For the patch format, I want it to provide step-by-step instructions.

- For the bot's personality, I want it to act like both an enthusiastic collaborator and a technical expert providing expert guidance.
- I will describe the sound in terms of descriptive words, genre context, and if necessary, I will provide a reference track or reference synth example from a song to help narrow down the sound/idea I'm going for.

From here, all that was left was to select the AI model that would power the bot and determine the knowledge base in which the bot would reference its outputs. I decided on Claude 4.5 Sonnet for the AI model, as I had used it for my previous chatbot and knew it provided the most dialogue-driven outputs with a conversational feel. I also uploaded the Serum 2 User Guide to the bot's knowledge base and ensured it would use this for all of its references to keep information as accurate and feasible as possible within Serum 2's interface. This first phase determined the configuration and instructions for how the bot would operate, which can be found in full in *Appendix I* of the Appendix section.

I began iterating and stress testing the bot at this point to determine its capabilities, accuracy, and dialogue style. The purpose of this was to identify any errors and issues that I could then go back and address within the bot's instructions. However, I was quite pleased with the bot's initial outputs from the start, and I was already beginning to see just how much potential this idea had.

Figure 3 shows the chatbot's interface, as well as the initial prompt used to test its capabilities:

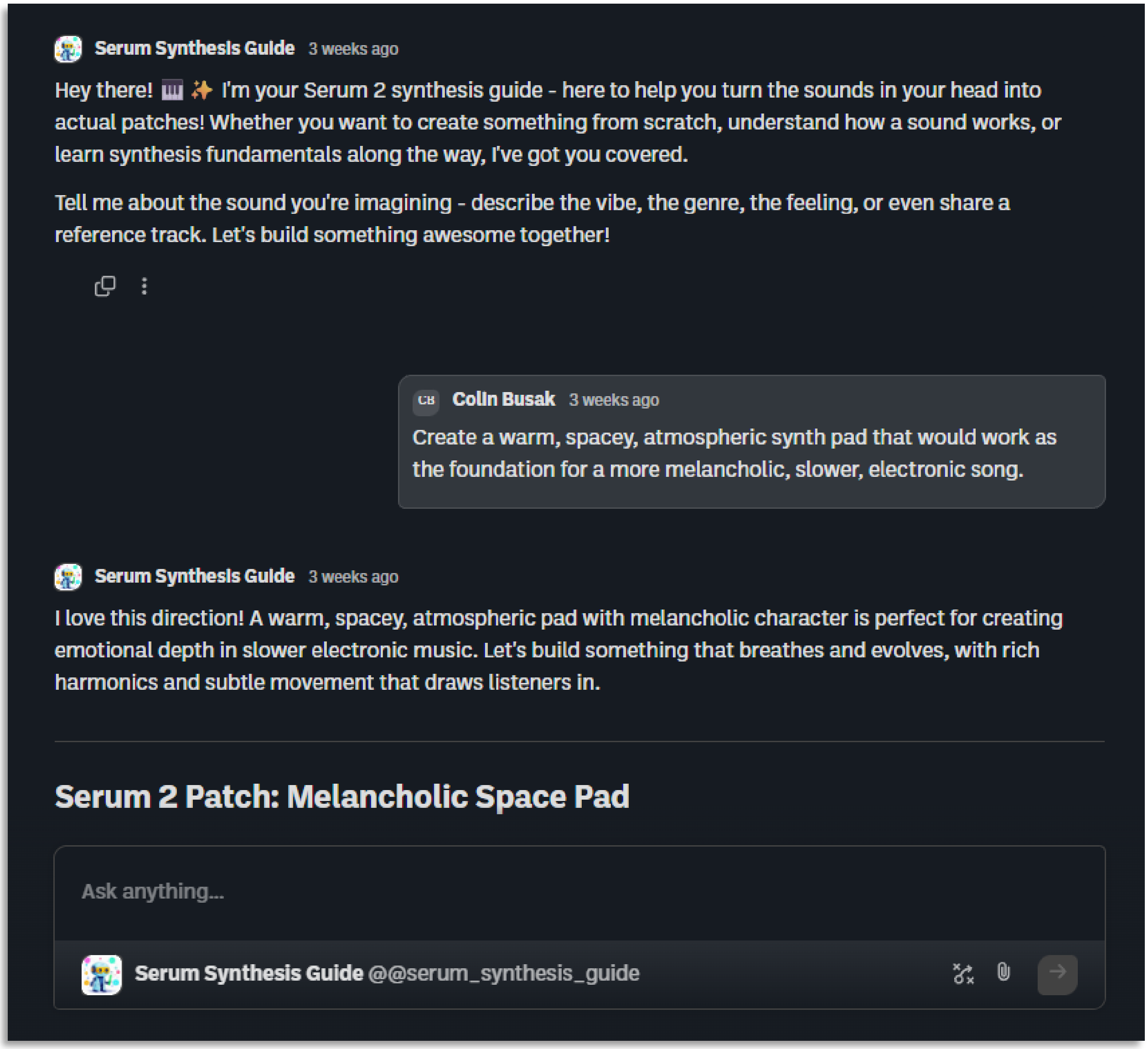


Figure 3: Serum Synthesis Guide's interface and my initial prompt

Just as I had hoped, the guide provided step by step instructions on how to create a spacey, melancholic synth pad patch from start to finish. It covered every parameter, effect, and modulation, while also providing context at each step as to how and why these decisions worked. However, when I began plugging this output into Serum 2 itself, I noticed that a few of the parameters and presets it had suggested for me were either under different labels in the actual interface or simply didn't work in the ways in which the guide was describing. I acknowledged this in a subsequent prompt and noted this in the bot's instructions. The full documentation of this chat can be found in *Appendix II* of the Appendix section.

Phase.8¿Interface.Pivot.and.Rhetorical.Consideration

After my initial idea was realized, I considered how I could expand on it both conceptually and rhetorically. I had already been through this process before with my previous chatbot and felt like I was reaching the ceiling within what BoodleBox could offer in my design. Additionally, I reflected on my own perspective and experience interacting with AI, which led me to the realization that someone like me might immediately reject the idea of interacting with a chatbot, an interface that many people now associate with AI generally, despite the intentions or functionality.

I went back to the Vibe Bot Builder to address this concern, where I had it create a fully standalone HTML file that would function as both a demo for the chatbot and a prototype for future iterations. The first iteration of the website was sleek, modern, and allowed users to try four example patch generations to get a sense for what the bot could do and what interaction with the bot looked like. While the functionality of the guide through this website was limited to these example patches, I felt like it showcased my idea in a much more user friendly and rhetorically appropriate way. Figure 4 shows the interface for Version I:

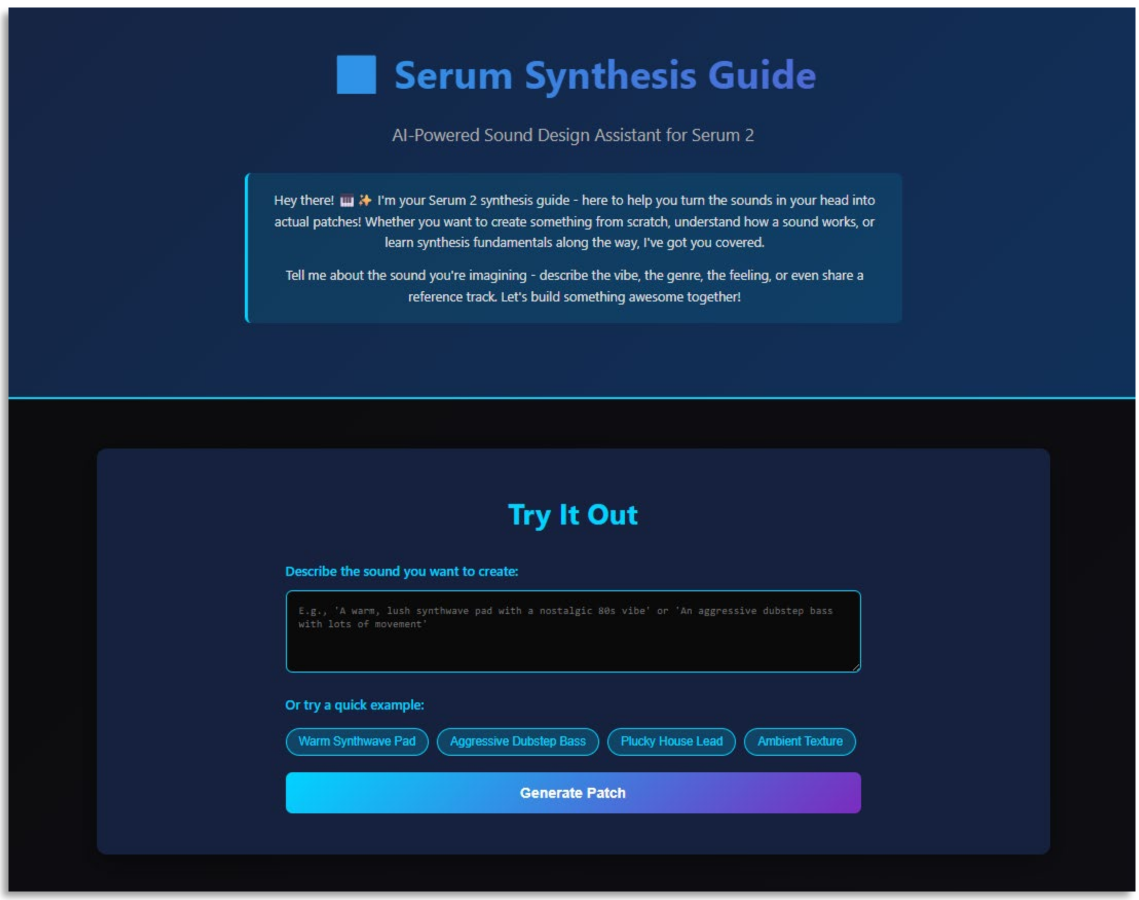


Figure 4: Serum Synthesis Guide Version I's Interface

Some notable opportunities for iteration included the website's overall color scheme and the unified, single-page layout in which all the background information could be found underneath the patch generator. To address these issues, I prompted the bot to create a new version with a color scheme more focused around Serum's iconic, neon green theme, separate tabs for each section to pivot away from the single-page design, and a light and dark mode feature for a more tailored user experience.

This new version felt close to a fully realized prototype. Figure 5 shows Version II's interface:

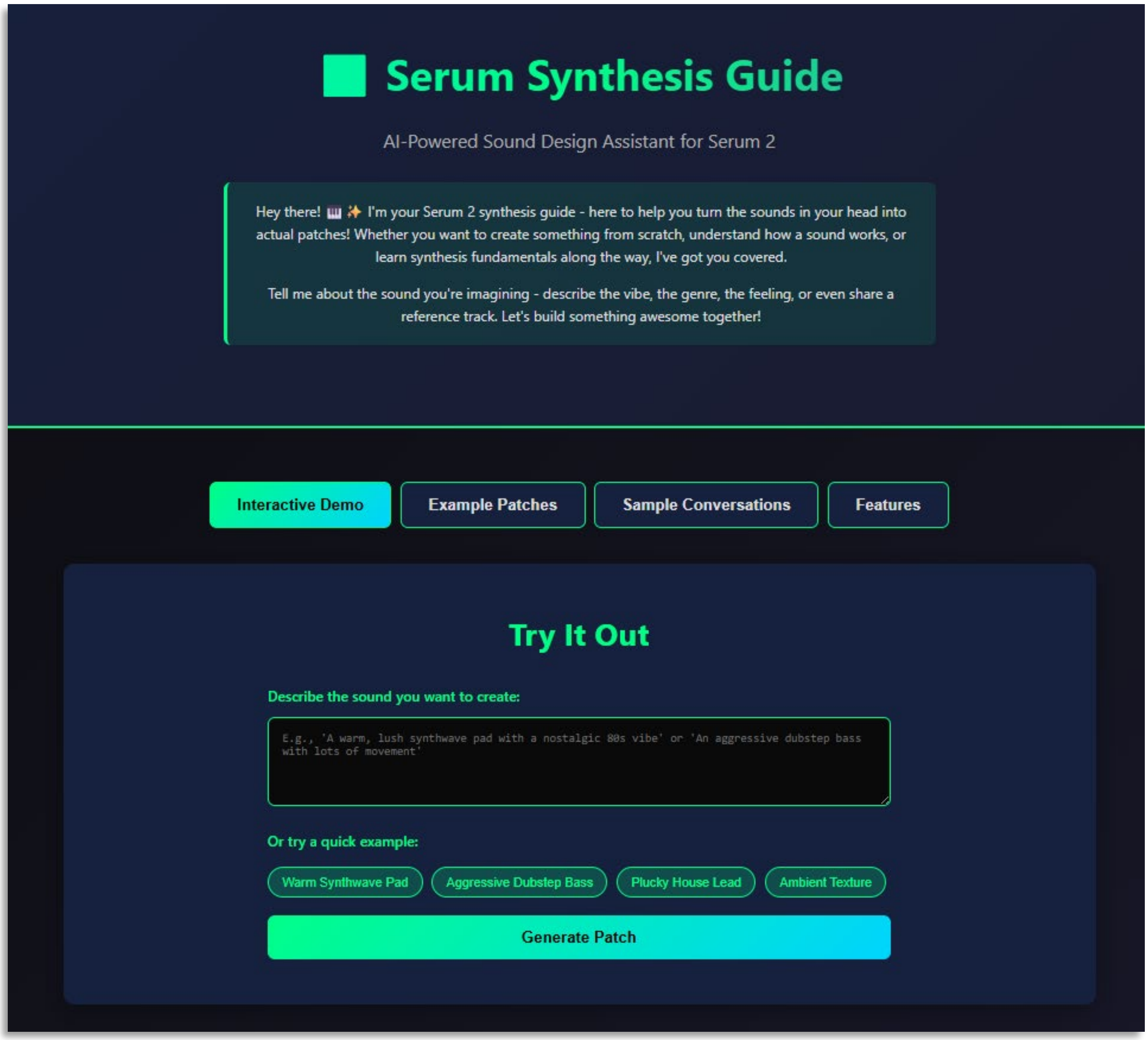


Figure 5: Serum Synthesis Guide Version II's Interface

Phase.9¿Implementing.Transparency.

For the final iteration of the Serum Synthesis Guide, I wanted to implement some kind of AI transparency section, which would provide the necessary acknowledgements regarding ethical AI use within this project. This was an integral aspect of the prototype and really brought everything full circle for me. From the very beginning of this class and this project, my goal was always to find a way to ethically integrate AI into a personal workflow and to explore opportunities in creative fields. Incorporating transparency into this prototype functions as the final pillar of its design.

Returning to BoodleBox's Vibe Bot Builder, I successfully implanted this section in another separate tab for ease of access. You can find a screenshot of the interface of the Serum Synthesis Guide's final iteration, as well as links to this version and the original chatbot with full functionality in the section titled, **Exhibit Feature: The Serum Synthesis Guide**, below.

Given more time, I would have liked to implement more accessibility features such as keyboard navigation, proper heading structure and ARIA features for screen readers, etc. Additionally, I believe even this iteration could be expanded upon further in its design. While it serves its purpose as a demo and prototype, a fully realized, future iteration would likely be a VST plugin that could integrate into any DAW software.

My Personal Knowledge, Growth, and Skill

Throughout TCID 4060, our course centered around three overarching concepts: AI, accessibility, and augmentation. Throughout each major project and particularly during the development of the Serum Synthesis Guide, I tried to incorporate all three themes. Ultimately, this was out of desire to challenge myself. Coming into this class and this project, I held and still do hold many hesitations regarding AI. However, over the course of this semester and given the current landscape of technology, I realized that AI and its integration into our everyday lives is inevitable. There is a real conversation that must be had regarding the future of AI, its impacts both culturally and environmentally, and how it can be used ethically.

Through our many discussions in class and our major assignments, I came to terms with this reality and while it seems daunting, I know now that it is possible. As students of technical communication, we reside in the intersection of technology, communication, and culture. We have the opportunity to work with emerging technology to determine how it can be improved for its users and future users. We can be the spearhead of change in the conversation of AI and how it augments our lives. Looking forward, it's clear that humanity and technology are slowly becoming intertwined and unified. We can certainly expect to see more kinds of augmentations in the future, be they cognitive like the Serum Synthesis Guide, physical in the form of wearables or prosthetics, or both.

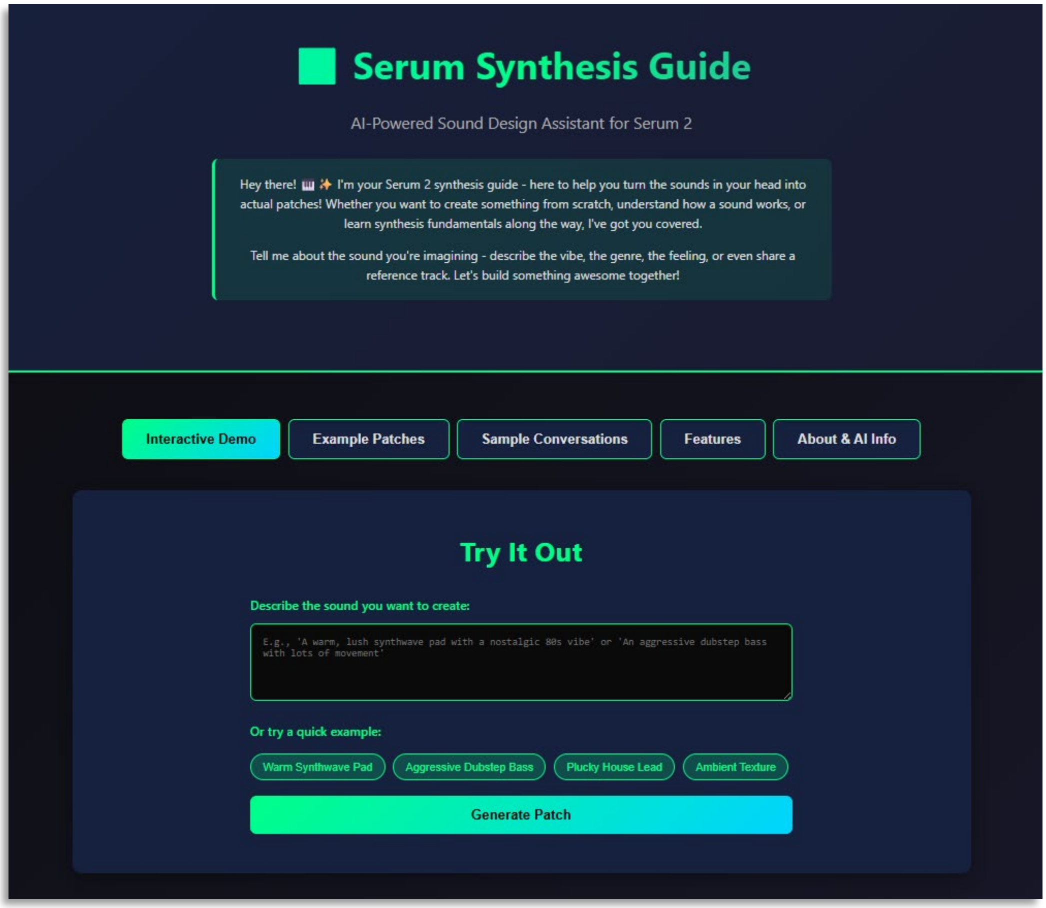
With that said, it is imperative that with all things AI or augmentative, we strongly emphasize accessibility for the user. This does not simply involve making things easier to understand or use. It also means making them ethical and responsible, so that we can continue to utilize and improve them for future generations to come. I believe there is an incredible amount of value in developing ideas that encourage users to interact with that which they do not understand. By bridging this gap, learning occurs in both the user in terms of what they need, want, and disapprove of, and the designer in how they can shape further iterations to better serve the user. In retrospect, this is the exact process I went through during TCID 4060 and while developing the Serum Synthesis Guide.

As this semester comes to a close, I want to thank Dr. Brandon Strubberg for facilitating this class. I have truly learned so much about myself, the three core tenets of the course, and what user experience means in the context of technology. I developed many skills that I will carry with me into later classes and careers. Though most of all, I received a glimpse into a future in which I can create change, lasting impact, and a better, more responsible and accessible experience for all.

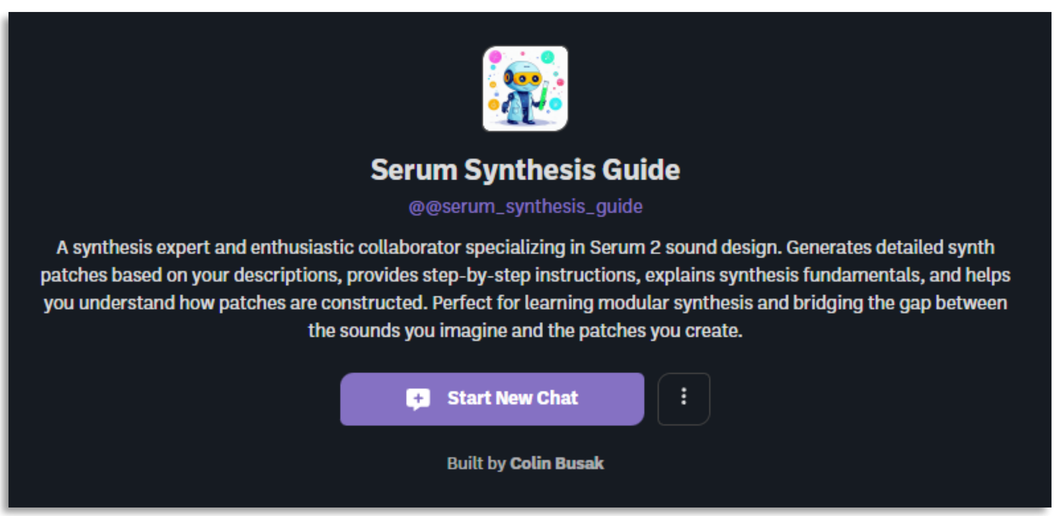
Finally, I would like to thank you, the reader, for considering my thoughts, ideas, designs, and creations that developed over the course of this semester. The following page acts as the exhibit feature for my prototype of the Serum Synthesis Guide, where you can find links to the website and the original chatbot with full functionality. Try them out for yourself and never stop pursuing your musical endeavors!

For inquiries, questions, or feedback, please contact me at the following email:
cbusak2@uccs.edu

Exhibit Feature: The Serum Synthesis Guide



Click here: Serum Synthesis Guide: Prototype Website



Click here: Serum Synthesis Guide: Chatbot

APPENDICES

Appendix I

[Serum Synthesis Guide's Configuration and Instructions]

Appendix II:

[Documentation of Chatlogs with Serum Synthesis Guide]